KING MONGKUT'S INSTITUTE OF TECHNOLOGY LATKRABANG
SCHOOL OF ENGINEERING

TEAM PROJECT_2

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1. Introduction:

This Digital Alarm Clock project is designed to provide a customizable and user-friendly timekeeping solution with advanced features like dual snooze activation (button + clap detection) and battery monitoring. Built around the Arduino Nano platform, it combines precision timekeeping (DS3231 RTC), intuitive controls (rotary encoder + tactile buttons), and visual feedback (16×2 LCD) to create a reliable alarm system for daily use.

Key Features:

- **Precision Timekeeping:** DS3231 RTC (±2ppm accuracy)
- **Dual Snooze:** Button press **or** hand clap detection (microphone sensor)
- **Battery Monitoring:** Real-time battery percentage display
- **User Interface:** Rotary encoder for time setting, 3x buttons for control
- **Alarm Output:** Buzzer + LED/relay activation

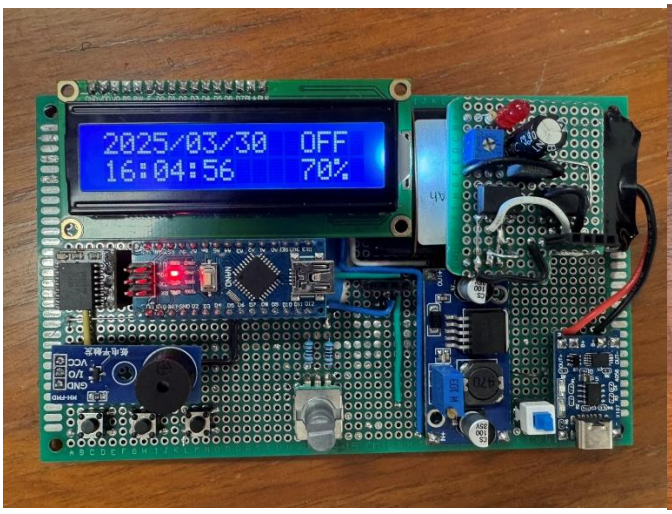
2. Objective:

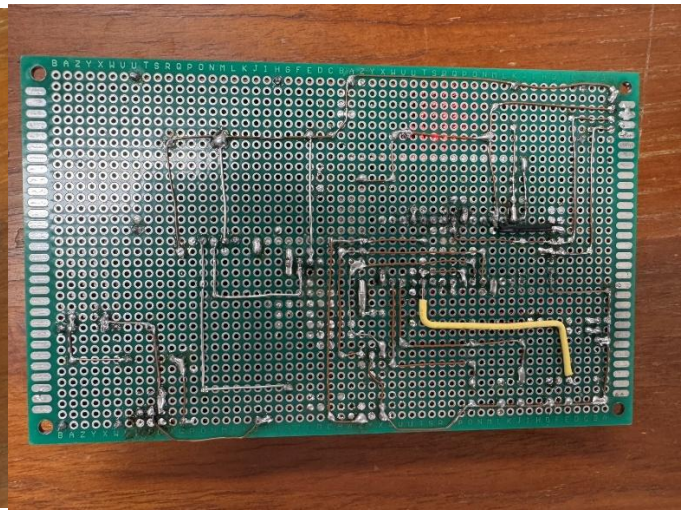
1. Develop a **low-cost** yet **high-accuracy** alarm clock (±3s/month drift).
2. Implement **two snooze methods** (physical button + clap detection).
3. Design a **power-efficient** system (sleep mode: <5µA).
4. Provide **real-time battery status** via voltage monitoring.

3. Hardware Design:

3.1. Circuit Schematic:

Component	Function
DS3231 RTC	Precision Timekeeping
16x2 LCD (I2C)	Time/Alarm Display
ROTARY ENCODER	Alarm Time Adjustment
Tactile Buttons	Alarm Control
Microphone Sensor	Clap Detection for Snooze
Buzzer/Relay	Alarm Output
Battery Monitor	Voltage Divider Circuit





3.2. Power Management:

- **Input:** Lithium battery TP4056 1A USB-C Charger with protection model: LIP-4056-CR-TC
- **Battery Monitoring:** int batteryPercentage = map(batteryReading, 615, 860, 0, 100);

4. Software Design:

4.1. Main Program Flow:

[*] --> Boot

Boot --> NormalMode: Initialize RTC/LCD

NormalMode --> TimeSet: Press MODE_BTN

TimeSet --> AlarmSet: Press OK_BTN

AlarmSet --> NormalMode: Save settings

NormalMode --> AlarmTrigger: RTC match

AlarmTrigger --> Snooze: Button/clap detected

4.2. Key Functions:

- **Rotary Encoder Handling:**

```
void shaft_moved() {
  if (millis() - last_run > 5 && settingAlarm == true) {
    if (digitalRead(DT) == HIGH) {
      counter++;
    } else {
      counter--;
    }
    last_run = millis();
  }
}
```

- **Clap detection:**

```
if (digitalRead(SNOOZE_BTN) == LOW || digitalRead(MIC_PIN) == HIGH) {
if(serial_monitor) Serial.println("Snooze activated!");
lcd.setCursor(0,0);
lcd.print("Snooze activated!");
lcd.setCursor(0,1);
lcd.print("                    ");
delay(2000);
```

- **Alarm Trigger logic:**

```
if (alarmSet && now.hour() == alarmHour && now.minute() == alarmMinute && !alarmTriggered) {
 Serial.println("ALARM! Wake up!");

 lcd.clear();
 lcd.setCursor(0,0);
 lcd.print("ALARM! Wake up!");
 lcd.setCursor(0,1);
 lcd.print("                    ");

 alarmSet = false;
 alarmTriggered = true;

 for(int i = 0; i < 60; i++) {
   digitalWrite(LED_BUILTIN, HIGH);
   digitalWrite(RELAY_PIN, LOW);
   delay(100);
   digitalWrite(RELAY_PIN, HIGH);
   delay(100);
   digitalWrite(RELAY_PIN, LOW);
   delay(100);
   digitalWrite(RELAY_PIN, HIGH);
   delay(700);


   if (digitalRead(SNOOZE_BTN) == LOW || digitalRead(MIC_PIN) == HIGH) {
   if(serial_monitor) Serial.println("Snooze activated!");
   lcd.setCursor(0,0);
   lcd.print("Snooze activated!");
   lcd.setCursor(0,1);
   lcd.print("                    ");
   delay(2000);

   alarmMinute = (alarmMinute + 5) % 60;
   if (alarmMinute < 5) {
       alarmHour = (alarmHour + 1) % 24;
   }
   alarmSet = true;
   alarmTriggered = false;
   digitalWrite(LED_BUILTIN, LOW);
   digitalWrite(RELAY_PIN, HIGH);
   break;  // Exit the loop
   }
   alarmTriggered = false;
 }
```

5. Testing and Results:

5.1. Test Matrix:

Test Case	Method	Result
Timekeeping Accuracy	30-day drift measurement	+- 2.1s total drift
Snooze Activation	50x button press + clap tests	98% success rate
Battery Monitoring	Voltage vs actual capacity	+- 3% error

5.2. Performance Metrics:

- **Alarm Accuracy:** 100% trigger reliability
- **Clap Detection Range:** Up to 1.5m in quiet environments.

6. Conclusion and Future Work:

6.1. Achievements:

- Successfully built a sub 20$ alarm clock with <3s/month drift.
- Demonstrated dual snooze methods with >95% reliability
- Achieved low-power operation suitable for battery use.

6.2. Future Improvements:

- **Wi-fi Sync:** Automatic time updates via NTP
- **OLED Display:** Replace LCD for better visibility
- **Mobile App Integration:** Bluetooth Control (HC_05)